

Methods: screening non-jointly-genotyped trio callsets for inherited rare variation with hprv

hprv pipeline — github.com/jlanej/high_priority_rare_variant

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A methods description derived from the pipeline code and shipped configuration (config/config.example.yaml) as of 2026-09-08 (the git history records the exact commit). Every threshold quoted is the shipped default and is configurable; the implementing script for each step is listed in Table 3. A rendered copy is at docs/methods.pdf.

1. Overview and design principles

hprv is a container-based pipeline that screens per-trio germline VCFs for high-priority inherited rare variants and consolidates genes in which rare functional variants recur across unrelated probands. It was written for the Gabriella Miller Kids First (GMKF) per-trio callsets: GRCh38 VCFs produced by the GATK genotype-refinement workflow, in which each trio was genotyped independently and the cohort was never jointly genotyped. That property shapes the design. Because a locus absent from one trio’s VCF cannot be read as homozygous reference, the pipeline never constructs a cohort genotype matrix and never derives allele frequency from internal counts; population frequency is taken exclusively from gnomAD v4.1, as a single quantity fixed for the whole run (Section 7). Cross-trio information is used only as a recurrence count of distinct individuals (Section 12) and, in the prioritization layer, as a quality signal (Section 13).

The analysis proceeds through a trio-resolution preflight and ten steps (Steps 0 to 9). Variant loci are unioned across trios into a site-only list (Step 1), annotated once (Step 2), reduced to biologically plausible sites (Step 3), and then the true per-trio genotypes are recovered at those sites (Step 4) and classified by inheritance mode with genotype-level quality control (Step 5), then re-scored for splicing at a wide window (Step 5b). Calls are consolidated per gene (Step 6), summarised for review (Steps 7 and 8), and re-ranked by a gene-level excess statistic, an artifact panel, and a per-variant evidence tier (Step 9).

Three rules govern every stage. First, rarity, functional impact, and genotype quality are gated before, and independently of, any gene list or constraint metric; gene lists and constraint act only as ranking priors, so a variant in a gene with no prior annotation is never removed (the “never-drop” rule). Second, one population-frequency quantity is used for every variant in a run; the run-level choice and the per-variant provenance are written into the output. Third, every step is idempotent with content-keyed completion markers, halts on an incoherent configuration or a transfer that matched nothing, and writes its input and output counts to a run audit (Section 15).

The scope is inherited germline single-nucleotide and small indel variation under dominant, recessive (homozygous and compound heterozygous in trans), X-linked and Y-linked models. De novo variants are detected only as a secondary cross-reference; mitochondrial contigs are excluded at the union step; copy-number and structural variants are not assessed.

2. Software environment and reproducibility

All tools run from one container image built on the Ensembl VEP release 115.0 image (`ensemblorg/ensembl-vep:release_115.0`), onto which a version-pinned micromamba environment is layered: `bcftools`, `samtools` and `htslib` 1.23, `bedtools` 2.31.1, and Python 3.11 with `cyvcf2` 0.31.1, `pysam`, `NumPy` 2, `pandas` 2, `SciPy`, `PyYAML` and `openpyxl`. The conda Perl is removed after the solve so that VEP and its plugins run under the base image's Perl. For the optional non-human-read screen (Section 14.3), `kraken2` v2.17.1 is compiled from source and the `nonhuman-screen` package is installed at a pinned commit (9e9eb6ce). An isolated conda environment holding the Illumina SpliceAI model (TensorFlow 2.13 to 2.15, Keras 2) supports the optional live splice-score backfill (Section 6.4). The pipeline is intended to run inside the image under Apptainer on HPC; the same scripts detect a host invocation and wrap each tool call in Apptainer or Docker.

A single YAML configuration file is the contract between the user and the pipeline. Filesystem locations are `{ENV}` placeholders resolved at run time, so no path, sample identifier, or controlled-access artifact is ever committed. The YAML is loaded and re-emitted as shell exports for the Bash steps, with booleans normalised to `true/false`.

The expensive steps (trio QC, the site union, annotation, per-trio genotype recovery, CRAM slicing and prioritization) write completion markers whose content is a key over the inputs that determine the output, and are skipped on re-run when the key matches; the inexpensive Python steps (selection, inheritance classification, gene consolidation and the workbook) recompute every run. The keys are: for the site union, the manifest plus each trio VCF's path, size and modification time, the `FILTER` expression, the excluded contigs and the sample subset; for annotation, a checksum of the union plus the identity of the cache and each plugin score file, the two transfer sources, the transcript selector, the frequency oracle, the VEP flag recipe and the list of lifted fields; for per-trio candidate VCFs, a checksum of the plausible-site file plus the source VCF identity and sample subset; for prioritization, a SHA-256 over every input table, the configuration and the scalar arguments. A marker whose key differs from the current inputs invalidates the cached product, and every bgzip product is integrity-checked (`bgzip -t`) before it is trusted.

At the start of Steps 3 and 5 the screen's parameters are validated and the run halts on any setting that would silently disable a filter: an unrecognised or empty impact list, a rarity ladder that is not ordered ($\text{dominant_max} \leq \text{recessive_max} \leq \text{benign_ba1}$, and $\text{recessive_strict} \leq \text{recessive_max}$), an allele-balance band outside $[0, 1]$ or with its lower bound at or above its upper bound, a de novo depth floor below the general depth floor, or a boolean given as a quoted string.

3. Inputs and trio resolution

The user supplies a tab-separated trios file and a VCF source (a directory searched recursively for `*.vcf.gz`, `*.vcf` and `*.bcf`, a file listing VCF paths, or both). The trios file must carry a header naming the proband, father and mother columns; names are matched case-insensitively against the aliases `kid/child/proband`, `dad/father/paternal` and `mom/mother/maternal`, with `_id`, `_sample_id` and `sample_` affixes tolerated. Column order is never assumed when a header is present, and a header that names some roles but not all three is an error rather than a positional guess, because a transposed mother and father would invert every parent-of-origin call while leaving the Mendelian-error rate unchanged. A file with no header is read positionally with a warning. Duplicate proband rows are skipped.

The trios file may also carry sex columns (`kid_sex`, with `sex`, `child_sex` and `proband_sex` accepted; `dad_sex`; `mom_sex`), whose values are 1/2/0 or male/female/unknown (m/f; a blank, `.` or NA cell is unknown). A stated proband sex is the canonical sex for that trio: it is written to the PED, Step 5 judges every

chrX and chrY decision under it, Step 6 counts male probands by it, and Step 0's chrX inference only checks it. A parent value that contradicts the role (a female father, a male mother) is an error, since it is the signature of a transposed pair and the pipeline does not guess which column is wrong; an unknown parent value defaults to the role. Any other spelling is an error rather than a silent unknown.

Sample membership is read from each VCF header only. For each trio, the candidate VCFs are those containing all three sample identifiers by exact match; a VCF may contain additional samples. When more than one VCF qualifies, the one with the fewest samples is chosen, with a lexical tie-break on path, and the multiplicity is recorded. A trio with no qualifying VCF is reported as unresolved in `trio_resolution.tsv` (naming the missing members) and skipped; the run fails only when no trio resolves. For each resolved trio a PED file is generated with the proband's sex as stated in the trios file (or unknown, 0, when no sex column was supplied, in which case Step 0's chrX inference fills it in downstream), the father coded male, the mother female, and the proband marked affected. `trio_resolution.tsv` records the sex written for each proband. The resolved manifest (`trios.resolved.tsv`: trio identifier, VCF, PED, comma-separated samples) is the input to Steps 0, 1 and 4, and the number of resolved trios is the screened-cohort size N used in Steps 6 and 9.

4. Per-trio quality control (Step 0)

Step 0 computes an advisory set of per-trio quality flags. No step excludes a flagged trio automatically; the flags are surfaced in the workbook and the review export for the curator.

For each trio, the VCF is scanned over autosomal biallelic records until 200,000 sites have passed the site-level QC below (`qc.max_sites`); on genome-wide data the statistics therefore describe a prefix of the genome, adequate for detecting sample swaps and gross contamination but blind to a localised Mendelian-error cluster beyond it. A site enters the Mendelian-error denominator when all three members are called and each has genotype quality (GQ) ≥ 20 and depth ≥ 10 . A violation is scored when a homozygous-alternate child has a homozygous-reference parent, a homozygous-reference child has a homozygous-alternate parent, or a heterozygous child has two homozygous-reference or two homozygous-alternate parents. The trio is flagged when the violation rate exceeds 0.02. The per-member no-call rate over the scanned biallelic autosomal records is also reported and flagged above 0.10, because a merged single-sample callset renders every non-carrier parent as missing rather than homozygous reference and would otherwise pass the Mendelian test with a cleaner-looking rate while defeating every parental test in Step 5.

Sex is inferred for all three members from chrX outside the pseudoautosomal regions (GRCh38 PAR1 chrX:10,001–2,781,479; PAR2 chrX:155,701,383–156,030,895). Among that individual's FILTER-passing (PASS or .), fully called biallelic chrX calls with $GQ \geq 20$ and depth ≥ 10 , a heterozygous fraction $\text{het}/(\text{het} + \text{hom-alt})$ below `qc.x_het_male_max` (default 0.10) is called male and otherwise female, provided at least 20 informative sites are available (else unknown). The genotype counted is the one the pre-refinement likelihoods favour where FORMAT/PL is present (and the GQ filter uses the likelihood-derived quality), because the genotype refinement's diploid pedigree prior pushes a male's rare hemizygous alternate calls to heterozygous (Section 11) and would otherwise inflate his heterozygous fraction. Filtered records and half-called genotypes are excluded because both are enriched for the artifacts that render a hemizygous male as heterozygous. A father inferred female or a mother inferred male raises `parent_sex_flag`; the accompanying warning distinguishes the pattern a transposition produces (father female and mother male) from a single mis-reading parent, which indicates a sample swap or a miscalibrated cutoff rather than transposed columns.

The cutoff is callset-dependent and the default is not appropriate for a diploid-called cohort: GATK genotypes chrX as diploid in males, so a hemizygous male's residual heterozygous calls place his fraction well above

0.10 (on one 221-trio GRCh38 WGS cohort the default inferred no father and one proband male). The report therefore carries each parent's raw fraction and site count (`dad_x_het_ratio`, `mom_x_het_ratio`), fathers being known males and mothers known females, and the audit records the median of each; the cutoff belongs in the gap between the two distributions. When more fathers read female than male, Step 0 warns that the cutoff, not the pedigree, is wrong.

A second, audit-only sex check reads chrY coverage rather than genotypes. Anchor sites are the FILTER-passing non-pseudoautosomal chrY records at which the father is a confident hemizygous alternate carrier on the pre-refinement likelihoods (FORMAT/PL; $\text{depth} \geq 10$, likelihood-derived $\text{GQ} \geq 20$, allele balance ≥ 0.90), the sites this callset demonstrably maps Y reads to in a male. At those sites the proband's and the mother's depth (missing read as zero) are recorded, and each member's median anchor depth, divided by that member's median autosomal depth, is expressed as a fraction of the father's (`y_cov_ratio`, `mom_y_cov_ratio`): a son reads close to one and a daughter close to zero. The proband is called male at a ratio ≥ 0.30 (`qc.y_cov_male_min`) and female at ≤ 0.10 (`qc.y_cov_female_max`), otherwise indeterminate, provided at least 50 anchors exist (`qc.y_min_anchor_sites`); the fraction of anchors at which the proband has ≥ 3 reads (`qc.y_reads_min_dp`) is reported as `y_covered_frac`. The mother serves as the in-trio female control: when her own ratio exceeds `qc.y_cov_female_max` no call is made for the trio (`y_flag=mother_y_coverage`). Because the statistic is a median over the anchors, a few mismapped reads in a female cannot produce a male call. The result is compared with the stated pedigree sex (`sex_match_y`) and with the chrX inference (`xy_agree`), each three-state as `sex_match` is, warned on any disagreement and recorded in the audit with the median ratio by pedigree sex and the mothers' median; it decides nothing, since the pedigree is canonical and the chrX inference is what fills in an unknown, and it does not enter `overall_pass`. Genotypes are deliberately not read for the proband or the mother on chrY, because after genotype refinement a chrY genotype is the diploid pedigree prior's opinion (Section 11). Beside the anchored statistic, each member's raw chrY coverage is reported: the median depth over every FILTER-passing non-pseudoautosomal chrY record, as a fraction of half that member's autosomal median depth (`kid_y_cov_haploid`, `dad_y_cov_haploid`, `mom_y_cov_haploid`), so that a male reads close to one and a female close to zero on the same scale. Under the same bands this reads both parents' sex against their roles (`dad_y_sex`, `mom_y_sex`, `parent_sex_flag_y`), the one parent check that involves no callset-dependent cutoff; a transposed father and mother read as `dad=2` and `mom=1`, and the warning names a transposition only for that pattern. When the father yields fewer than `qc.y_min_anchor_sites` anchors, the proband is called on this raw ratio instead (`y_basis=raw`), with the mother's raw ratio as the control.

The proband's sex is resolved by one precedence rule shared by Steps 0, 5 and 6: a sex stated in the trios file is canonical, and the inference fills in only when the pedigree leaves it unknown. `sex_source` records which applied (ped, inferred or none). `sex_match` compares the two when both exist: 1 when they agree, 0 when they disagree, blank when no comparison was possible. A disagreement is reported (the trio fails `overall_pass`, a warning is issued, and Step 5 flags every call of the trio `sex_discordant_inference`) but never resolved by overriding the pedigree.

Contamination is assessed from verifyBamID FREEMIX when a directory of `.selfSM` files is configured (flag above 0.05). Otherwise a VCF-only proxy is used: for each member, the summed reference-allele depth divided by the summed total depth over that member's homozygous-alternate SNV sites with $\text{GQ} \geq 20$ and $\text{depth} \geq 10$ (flag above 0.02). This proxy is uncorrected for allele frequency and reference bias and detects only gross contamination; the calibrated CHARR estimator is not implemented. `overall_pass` is set when no flag fired.

5. Cohort site union (Step 1)

For each resolved trio the source VCF was reduced to its three members (`bcftools view -s`), restricted to records with FILTER PASS or ., and stripped of the mitochondrial contigs (`chrM`, `chrMT`, `M`, `MT`). Multiallelic records were split and indels left-aligned against the reference (`bcftools norm -m- -f`, with a reference mismatch treated as fatal), alleles no member carries after splitting were removed (`--min-ac 1`), genotypes were dropped (`view -G`) and all INFO fields were removed. The per-trio site files were then concatenated with duplicate removal (`bcftools concat -a -D`), sorted, and collapsed to unique representations (`norm -d exact -f`), yielding one site-only cohort union. The union is verified to contain no record on an excluded contig. Mitochondrial exclusion is enforced here rather than left unmodelled because every inheritance model downstream is diploid, and homoplasmic rCRS-referenced haplogroup variants, which carry no gnomAD frequency, would otherwise be called as homozygous recessive in every trio.

6. Variant annotation (Step 2)

6.1 Ensembl VEP

The cohort union was annotated once, never per trio, with Ensembl VEP release 115 against the matching offline GRCh38 cache (`--cache --offline --assembly GRCh38 --fasta`), sharded by contig for resumability and re-assembled before any downstream operation, so the product is identical to a single pass. VEP was run with `--symbol --biotype --numbers --total_length --hgvs --canonical --mane --af_gnomade --af_gnomadg --max_af --check_existing --flag_pick --pick_order mane_select,mane_plus_clinical,canonical,rank`. The gnomAD v4.1 per-population exome and genome allele frequencies and the ClinVar clinical significance therefore come from the VEP cache itself. Six plugins were applied. Ensembl's stock NMD plugin, which needs no data file, marks a stop-gain, frameshift or canonical splice variant predicted to escape nonsense-mediated decay (last exon, within 50 nt of the penultimate exon's end, first 100 coding bases, or an intronless transcript). The four score plugins were CADD v1.7 (the whole-genome SNV table and the gnomAD-genomes indel table), SpliceAI (the precomputed raw genome-wide SNV and indel delta-score files, Illumina `genome_scores_v1.3`; the plugin's own cutoff is deliberately not set so raw scores are carried), REVEL v1.3 (the GRCh38-sorted distribution), and AlphaMissense (`AlphaMissense_hg38.tsv.gz`, with `transcript_match=1` so a score is taken only from the transcript VEP annotated). The sixth, SpliceVault (Dawes et al. 2023), is optional and adds empirical rather than predicted evidence: for a variant SpliceAI predicts to disrupt an annotated splice site, the plugin reports the most frequent natural mis-splicing events observed at that site across the 300K-RNA compendium of roughly 335,000 RNA-seq samples — skipping of one or more exons, or use of a cryptic donor or acceptor at a stated offset — each with its reading-frame consequence and the percentage of supporting samples, together with the site's sample count, maximum depth and the fraction of the top events that shift the frame. The plugin is a review resource: when its table is absent Step 2 warns and the corresponding columns are left blank, and no selection or ranking step reads them.

Because `--flag_pick` retains every consequence block and marks VEP's choice with `PICK=1`, the consequence used downstream is selected when the CSQ fields are lifted to INFO with `bcftools +split-vep`: the picked block when a `PICK` field exists, otherwise the worst consequence (`resources.vep.csq_select` overrides this). The lifted fields, each prefixed `vep_`, are Consequence, IMPACT, SYMBOL, Gene, Feature, BIOTYPE, EXON, INTRON, HGVSc, HGVSp, cDNA_position, CDS_position, Protein_position (each with its total), MANE_SELECT, NMD, CADD_PHRED, CLIN_SIG, gnomADe_AF, gnomADg_AF, MAX_AF, MAX_AF_POPS, the ten grpmax-eligible population frequencies (gnomADe and gnomADg for AFR, AMR, EAS, NFE and SAS), the SpliceAI delta scores and positions (`DS_AG`, `DS_AL`, `DS_DG`,

DS_DL, DP_AG, DP_AL, DP_DG, DP_DL, SYMBOL), REVEL, am_pathogenicity and am_class, STRAND, and the seven SpliceVault fields (top_events, out_of_frame_events, site_pos, site_type, site_sample_count, site_max_depth and SpliceAI_delta).

The step halts if Consequence, IMPACT or SYMBOL is missing, if none of the ten grpmx-eligible frequency fields is present, if SpliceAI or CADD is configured but not one site received a score, or if no site carries any gnomAD frequency value; each of these would otherwise leave a run that exits successfully with a silently disabled gate.

6.2 The two bcftools annotate transfers

Exactly two annotations are transferred from external VCFs, both because the VEP cache cannot supply them. From the ClinVar GRCh38 sites VCF (pinned release 2026-07-06, contigs renamed to the chr prefix during preparation), CLNREVSTAT and CLNSIG are transferred as `clinvar_CLNREVSTAT` and `clinvar_CLNSIG`; the review status is later mapped to gold stars (practice guideline 4, expert panel 3, multiple submitters without conflict 2, single submitter or conflicting 1, no assertion criteria 0). From a slim of the gnomAD v4.1 joint (exomes plus genomes) sites VCF, prepared by retaining five INFO fields (AF_joint, AF_grpmx_joint, fafmax_faf95_max_joint, fafmax_faf95_max_gen_anc_joint, nhomalt_joint), those fields are transferred as `gnomad_AF_joint`, `gnomad_AF_grpmx`, `gnomad_faf95`, `gnomad_faf95_group` and `gnomad_nhomalt`. Both transfers are allele-exact (chromosome, position, reference and alternate) and each is guarded: a transfer that matched zero sites is fatal, because a cohort union always overlaps both resources and zero matches indicate a build or contig-naming mismatch. The gnomAD transfer is required under the default frequency oracle (Section 7); the ClinVar transfer is optional and its absence leaves the review status “unavailable”, which is distinguished from zero stars everywhere downstream.

6.3 Ingesting an external VEP VCF

A pre-existing VEP output may be supplied in place of running VEP. It must carry a `##VEP` header and must not declare a non-GRCh38 assembly; it must contain no multiallelic records, and it must cover every site of the cohort union (uncovered sites would be silently lost at Step 4), unless a partial ingest is explicitly permitted and the shortfall recorded. The same field lifting, guards and transfers then apply.

6.4 Optional live SpliceAI backfill (Step 2b)

Off by default. When enabled, variants in the annotated union with no precomputed SpliceAI score (indels only, by default, because the precomputed SNV set is genome-wide) are scored with the Illumina SpliceAI model (`-A grch38 -D 500`) in the isolated environment. For each variant the maximum of each delta score over the genes reported is written into the same `vep_SpliceAI_pred_DS_*` fields, so Step 3 reads precomputed and backfilled scores identically. An unavailable environment halts the run; a transient scoring failure leaves the union on precomputed scores.

7. Population allele frequency: one oracle per run

All rarity decisions read one function, `annotations.frequency()`, which returns a single quantity selected for the whole run by `resources.gnomad.oracle`.

Under the default, `faf95`, the value is gnomAD’s filtering allele frequency `fafmax_faf95_max_joint`: the lower bound of the 95% Poisson confidence interval on the allele frequency, maximised over the FAF-eligible

genetic-ancestry groups (afr, amr, eas, mid, nfe, sas), which is the quantity the ACMG/AMP and ClinGen frequency-filtering recommendations specify. gnomAD emits this field as missing, not zero, wherever no group's interval lower bound exceeds zero. The pipeline therefore resolves an allele in three ways and records which applied in `rarity_basis`: a published `faf95` is used as is (measured); an allele present in the gnomAD record but with no `faf95` resolves to 0.0 (`zero_ci`, the rarest value); and an allele with no gnomAD record at all resolves to missing (`absent`), which every gate also treats as rarest. The witness separating the last two cases is the transferred `gnomad_AF_joint` field. The VEP-cache point estimates are never consulted on this arm.

Under the explicit opt-down, `grpmax_proxy`, the value is the maximum of the VEP-cache point estimates over the `grpmax`-eligible groups only (AFR, AMR, EAS, NFE, SAS, exome and genome), which mirrors gnomAD's own exclusion of the bottlenecked `ami`, `asj`, `fin` and `mid` groups. The cache carries no allele counts, so no interval correction is possible; the proxy sits about one interval width above `faf95` for low-count alleles and errs toward removing them. Cache frequencies exist only for dbSNP-accessioned alleles, so an unaccessioned gnomAD allele reads as absent under this arm.

VEP's `MAX_AF` (which maximises over founder and 1000 Genomes populations) and the global gnomAD frequencies are carried for reporting only and are not filter fields under either arm. Where a field holds several values (multiple consequence blocks), the maximum is taken. Step 5 resolves the value once per call and writes `rarity_af`, `rarity_oracle` and `rarity_basis`; Steps 6 and 9 consume those columns rather than re-deriving them. Under `faf95`, Step 5 also counts calls with no gnomAD record that nevertheless carry a cache frequency; because the joint release is a superset of the cache, a non-zero count indicates an under-matched transfer and is reported.

8. Selection of plausible sites (Step 3)

Step 3 is an inheritance-agnostic reduction of the annotated union applied in a fixed order. A site with frequency ≥ 0.05 (the ClinGen BA1 threshold) is removed and never rescued, including by a ClinVar assertion. A site is otherwise retained on rarity if its frequency is missing, is below the permissive recessive cutoff 1×10^{-2} , or the site carries a ClinVar pathogenic or likely-pathogenic assertion (`CLIN_SIG` containing "pathogenic" without "conflicting" or a benign qualifier; review status is deliberately not consulted). A pathogenic or likely-pathogenic site is then kept outright. Every other rare site must pass a three-rung functional ladder, any rung sufficing, evaluated in order: VEP IMPACT of HIGH or MODERATE; maximum SpliceAI delta score (over acceptor and donor gain and loss) ≥ 0.2 ; CADD PHRED ≥ 25.3 . A missing SpliceAI or CADD score never removes a site; it merely fails to retain it. Each retained site is tagged with its keep reason (`hprv_keep_reason`, one of `clinvar_plp`, `impact_high`, `impact_moderate`, `spliceai`, `cadd`), and the audit records the drop reasons (`ba1`, `too_common`, `not_functional`), splitting `not_functional` into sites both predictors scored and rejected versus sites neither scored.

Two consequences of the ladder are worth stating. Every missense variant has IMPACT MODERATE and is retained at the first rung, so REVEL and AlphaMissense cannot change which variants pass the screen; they act only in Step 9. Conversely, CADD 25.3, a cutoff calibrated for missense, is in practice applied only to variants below MODERATE impact (intronic, synonymous, UTR and regulatory), where it functions as a discovery rank rather than as calibrated evidence. Gene lists and constraint play no part in this step.

9. Per-trio genotype recovery (Step 4)

The true genotypes of each trio at the plausible sites were recovered from the source VCF rather than from any merged product. The plausible loci were padded by 1,000 bp and merged into a BED, and each trio VCF was restricted to those regions and its three members, normalised as in Step 1 (`norm -m- -f`, this time warning rather than failing on a reference mismatch) with `--keep-sum AD` where an AD field exists, so that after splitting a multiallelic heterozygous genotype each allele’s reference depth includes the reads supporting the other alternate allele and allele balance remains interpretable. Alleles the trio does not carry were removed, and all INFO was stripped except the GATK `hiConfDeNovo` and `loConfDeNovo` tags. The result was intersected allele-exactly with the plausible set (`bcftools isec -c none -n=2 -w1`) and the Step 2 annotations were transferred onto it (`bcftools annotate -c INFO`) from a copy of the plausible set with the de novo tags removed, so the transfer can never overwrite a trio’s own de novo tag. The transfer is verified per trio: every candidate record must carry `hprv_keep_reason`, otherwise the step halts.

10. Genotype quality control

Genotype-level QC is evaluated on the refined genotypes with `cyvcf2` in strict mode, so that a half-called genotype (`0/.`) is treated as missing rather than homozygous reference. GQ is read from `FORMAT/GQ`, which in genotype-refined VCFs is derived from the posterior probabilities; depth is the sum of allelic depths when AD is present, otherwise `FORMAT/DP`; allele balance (AB) is the alternate-allele fraction of AD.

Five predicates are used (defaults from `filters.genotype_qc` and `filters.denovo`). A heterozygous carrier requires $GQ \geq 20$, $depth \geq 10$ and AB within $[0.25, 0.75]$. A homozygous-alternate carrier requires $GQ \geq 20$, $depth \geq 10$ and $AB \geq 0.90$. A de novo child requires $GQ \geq 20$, $depth \geq 20$ (≥ 10 on a hemizygous site, `denovo_min_dp_hemizygous`, since a male’s single X or Y carries half the autosomal depth) and AB within $[0.25, 0.75]$. A homozygous-reference non-carrier requires $GQ \geq 20$, $depth \geq 10$ and $AB \leq 0.10$. A “clean” parent requires $GQ \geq 20$, $depth \geq 10$, at most one alternate-supporting read and $AB \leq 0.10$. The carrier predicates fail when AD is absent; the two non-carrier predicates pass when AD is absent, because a GATK reference-block genotype (`GT:DP:GQ:MIN_DP:PL`, no AD) is the ordinary shape of a non-carrier parent and failing it would discard every such trio. Such a pass is marked (`parent_ad_unmeasured`, `trans_evidence_unmeasured`) so that a phase or de novo call resting on a genotype call alone is distinguishable from one resting on read counts. Records with a FILTER value other than PASS or . are skipped.

11. Inheritance-mode classification (Step 5)

Each trio’s candidate VCF is screened with the proband’s sex taken from the PED (the trios file’s stated sex) or, only when unknown there, from Step 0’s inference; a stated sex that the inference disagrees with is kept, and every call of the trio carries the flag `sex_discordant_inference`. Each call records the sex it was judged under (`child_sex`) and its provenance (`child_sex_source`: `ped`, `inferred` or `none`). If the sex remains unknown, non-pseudoautosomal X and Y records are skipped for that trio with a warning and autosomal modes still run. Non-PAR chrY records in a female proband are skipped. A “rare” test at limit L passes when the oracle value is missing or below L ; the limits are 1×10^{-4} for the dominant and de novo models (`dominant_max`) and 1×10^{-2} for the recessive and X-linked models (`recessive_max`), with recessive and X-linked calls below 1×10^{-3} additionally flagged `high_conf_rarity`. Every emitted row carries the genotype of all three members as allele strings, the child’s GQ, depth and AB, the curated annotations (the HGVS coding and protein descriptions, exon and intron numbering, CDS position and MANE status of the transcript selected in Step 2, the NMD verdict of Section 13.4, the SpliceAI event decomposition below and,

when the SpliceVault table was configured, the empirical splicing outcome beside it) and a `flags` field; after the curated columns, every INFO field declared in the per-trio candidate VCF header is written verbatim under an `info_` prefix (the union over trios, in header order), including the complete VEP consequence string for every transcript, so no annotation computed in Step 2 is lost in the projection from VCF to table.

De novo (secondary). An autosomal de novo requires a heterozygous child with both parents homozygous reference, the child passing the de novo predicate, both parents passing the clean-parent predicate, and rarity at 1×10^{-4} . A male X-hemizygous de novo requires a homozygous-alternate son with a homozygous-reference mother passing the clean-parent predicate (the father's X is not transmitted to a son and is not consulted), the son passing the homozygous predicate with depth ≥ 10 (the hemizygous floor), and the same rarity. When the callset header declares `hiConfDeNovo`, the call additionally requires that tag to list this child (`filters.denovo.use_hiconf_tag`); when the header lacks the tag, detection rests on genotypes and QC alone. De novo rows carry `review_prior_crosscheck`, a reminder that genotype refinement with population priors can suppress an ultra-rare call, and are never the pipeline's primary result.

Homozygous recessive. An autosomal homozygous-alternate child passing the homozygous predicate, at rarity 1×10^{-2} . The parents grade the support of the call and never veto it: with both parents carrying the allele (heterozygous or homozygous) and each passing the predicate matching its own genotype, the row carries no flag; a carrier parent failing its own predicate is flagged `transmitting_parent_qc_fail`; a parent that is confidently homozygous reference (allele balance ≤ 0.10 at depth ≥ 10 and $GQ \geq 20$; `parent_ad_unmeasured` added when the call rests on no allele depths) is flagged `noncarrier_parent=mat` or `pat`, the shape of a hemizygous deletion in trans, uniparental disomy or a parental allele dropout, which the genotypes alone cannot distinguish; a parent that is uncalled, or homozygous reference with alternate reads above that band, is flagged `parent_gt_uninformative=mat` or `pat`, with `mother_alt_reads=N` or `father_alt_reads=N` when it has any. Each of those shapes was previously a silent no-row; the child's homozygous genotype is the evidence, and Step 6 counts carriers that rest on an unsupported parent apart.

X-linked recessive. For a male proband, a homozygous (hemizygous) alternate call at a non-PAR chrX site passing the hemizygous predicate, at rarity 1×10^{-2} . The mother grades the support as above: a carrier (or a carrier failing her own predicate, flagged) is the inherited shape; a mother who is confidently clean (homozygous reference, at most one alternate read, allele balance ≤ 0.10 , depth ≥ 10) is the de novo shape, handled by the secondary de novo model below with its own gates; a mother who is homozygous reference but not clean, or uncalled, leaves the origin unestablished and the son's call stands, flagged `parent_gt_uninformative=mat` or `noncarrier_parent=mat`, since a hemizygous variant in an affected boy is causally self-sufficient and the maternal genotype only separates inherited from de novo. A father carrying the allele does not veto the call but is flagged (`father_carries_x_allele`). On a male proband's non-pseudoautosomal chrX every member's genotype class is the one the pre-refinement likelihoods favour (FORMAT/PL; the refined genotype where PL is absent or flat, a refined no-call kept as a no-call) and the genotype QC uses the likelihood-derived GQ, for the reason given under the Y-linked model below: the father is haploid on chrX, so the refinement's diploid pedigree prior treats a son's hemizygous alternate with a hemizygous-reference father as a de novo event and repairs it by pushing the son to heterozygous, or by imputing a heterozygous call into the parent with the fewest reads. Whichever member's refined genotype was overruled is flagged (`child_gt_refined_discordant`, `father_gt_refined_discordant`, `mother_gt_refined_discordant`), and the `hiConfDeNovo` tag, computed on the refined genotypes, is reported but not consulted for the hemizygous de novo. A female proband's chrX is diploid and is judged on the refined genotypes, except her father's, which is haploid and is read from the likelihoods. For a female proband, a homozygous-alternate call passing the homozygous predicate at the same rarity, with her mother graded as a diploid parent and her father as a hemizygous one (a father who is hemizygous reference is `noncarrier_parent=pat`, the same deletion or dropout shape as an autosomal non-carrier parent); in Step

6 such a daughter is a biallelic hit and is tested under the biallelic null rather than the hemizygous-male one.

Y-linked. For a male proband, non-pseudoautosomal chrY records are judged on the pre-refinement likelihoods rather than on the refined genotype. GATK's genotype refinement (CalculateGenotypePosteriors) rewrites GT and GQ from a posterior that combines the caller's likelihoods with a diploid Mendelian pedigree prior and a population prior, and leaves FORMAT/PL as the caller produced it. On a haploid chromosome that prior is invalid, and on a refined GMKF trio its effects are visible: the mother is imputed a heterozygous genotype she has no reads for at father-son homozygous-alternate sites, and where her own mismatched reads confidently read homozygous reference the son's all-alternate call is pushed to heterozygous, since a homozygous-alternate child of a homozygous-reference mother is a de novo event under the prior. The hemizygous call therefore takes the genotype the likelihoods favour (genotype.hemi_call: the PL minimum among 0/0, 0/1 and 1/1, read as reference, mixed or alternate; the refined genotype only when no PL is present or the likelihoods are flat; a refined no-call stays a no-call), with GQ derived from the PL. A son who is hemizygous alternate (depth ≥ 10 , likelihood GQ ≥ 20 , allele balance ≥ 0.90) with a father who is hemizygous alternate is emitted as y_linked with origin=pat, gated at 1×10^{-2} like the X-linked model and flagged high_conf_rarity below 1×10^{-3} ; a father who is alternate but fails his own QC, or whose reads support both alleles, does not veto the call but flags it (transmitting_parent_qc_fail, and y_site_mixed_reads for the latter, the signature of a paralogous site). A son who is hemizygous alternate with a father who is confidently reference (depth ≥ 10 , at most one alternate read, allele balance ≤ 0.10) is emitted as denovo_y_hemi, a secondary de novo call requiring child depth ≥ 10 (denovo_min_dp_hemizygous, half the diploid floor because a haploid chromosome carries half the depth) and rarity at 1×10^{-4} ; GATK's hiConfDeNovo tag is reported but not consulted on chrY, because it is computed on the refined diploid genotypes, the imputed mother included. The mother's genotype is never read: she has no Y, so there is no maternal transmission and no Mendelian test. Her depth is recorded (mother_dp), and a mother with at least 3 reads (qc.y_reads_min_dp) at a male-specific chrY site flags the row y_female_reads, since reads in a female at such a site are X-derived mismatching and the son's allele there may be a mismatched maternal X allele. A son whose reads support both alleles is never a call (male_y_het); a father who is uncalled, or reference but not clean, leaves the sole transmitter unobserved and the son's call stands, flagged parent_gt_uninformative=pat (with his alternate reads) or noncarrier_parent=pat. Every row carries the likelihood-favoured genotype of all three members (child_gt_pl, mother_gt_pl, father_gt_pl) beside the refined ones, and a chrY row whose refined genotype the likelihoods overruled carries child_gt_refined_discordant or father_gt_refined_discordant. Unlocalised chrY scaffolds are treated as chrY; pseudoautosomal chrY positions, where present, are routed through the autosomal models.

Heterozygous collection and parent of origin. Every heterozygous child call outside male hemizygous regions that passes the heterozygous predicate, is rare at 1×10^{-2} , and has a gene annotation is pooled per gene with a parent-of-origin assignment. With both parents homozygous reference the origin is "de novo" only if both pass the clean-parent predicate. When exactly one parent carries the allele, the origin is that parent (maternal or paternal); the non-transmitting parent must be an affirmative, QC-passing homozygous reference for the origin to count as verified, otherwise the call is flagged origin_unverified; a verified non-transmitting parent with no allelic depth is flagged as a vacuous pass. When both parents carry the allele, a homozygous parent transmits obligately, so a heterozygous child of a homozygous and a heterozygous parent is assigned to the homozygous parent; two heterozygous parents give the indeterminate origin "both"; two homozygous parents cannot produce a heterozygous child and the record is counted as Mendelian-inconsistent. If the transmitting parent fails its own genotype QC the call is retained and flagged transmitting_parent_qc_fail rather than removed. A parent no-call leaves the origin unestablished.

Compound heterozygous. Within each gene, trans pairs are formed between maternal and paternal hets, and between a de novo het and a het of either parental origin; hets of origin "both" are never paired. Each

pair receives a `pair_id` and both legs are emitted as `compound_het`. A pair with a de novo leg cannot be phased from trio genotypes and is flagged `unphased_denovo_partner`; a pair whose trans evidence failed is flagged `origin_unverified`; a pair with a QC-failed transmitting parent is flagged `transmitting_parent_qc_fail`; a pair whose trans evidence rested on a parent without allelic depth is flagged `trans_evidence_unmeasured`. Only a pair carrying none of the first three flags is treated as a confirmed biallelic hit and consumes its two legs from the dominant model below.

Dominant. Every pooled het of maternal, paternal or indeterminate (“both”) origin that was not consumed by a confirmed compound-heterozygous pair and is rare at 1×10^{-4} is emitted as dominant with `origin=mat|pat|both` and any of the flags above. This inherited heterozygous class is the recurrence signal Step 6 consolidates. A variant may appear under more than one mode.

Accounting. Every record examined is classified as skipped (FILTER, unresolved sex, female chrY), called, or assigned a no-row reason (child not a carrier, a male Y het, a male X het, a child depth or quality failure, a child allele balance below or above the band — the postzygotic-mosaic or contamination shape and the allelic-imbalance shape, counted apart — a parent whose depth or quality fell short, a parent with alternate reads under a de novo-shaped het — the parental-mosaic or contamination shape, counted apart — rarity, no gene, a heterozygous child of two homozygous-alternate parents (the one remaining Mendelian inconsistency), a parental no-call under a heterozygous child, the inert [1×10^{-4} , 1×10^{-2}) heterozygous band, a disabled mode, or the hiConf tag), so that per trio the number examined equals skipped plus called plus no-row. ClinVar P/LP alleles the child carried that yielded no row are counted separately.

Splice event decomposition. The screen gates on the maximum of SpliceAI’s four delta scores; each call also carries the event behind that maximum. The component with the largest delta score at or above the screen’s 0.2 floor is named (`spliceai_event`: acceptor or donor, gain or loss) with the genomic position of the affected site (the variant position plus the model’s offset), and a second component at or above the floor is named alongside it. A gain and a loss of the same site type are read as a cryptic-site shift whose exon-boundary displacement is the difference of the two offsets and whose frame follows from that displacement modulo three; a lone loss is recorded as a site loss (exon skipping or intron retention, frame undetermined without the exon length), a lone gain as a site gain, two gains as a pseudoexon candidate, two losses as a whole-exon loss candidate, and mixed site types as complex (`spliceai_effect`). A call whose SpliceAI gene symbol differs from VEP’s picked gene is flagged.

Empirical splicing outcome (SpliceVault). Where the SpliceVault plugin annotated the call, the ranked 300K-RNA events are carried verbatim (`splicevault_top_events`) with the fraction that shift the frame, the site type the events are keyed to and its sample support, and the rank-1 event is named in a compact form (ES for exon skipping; CD+12 or CA-31 for a cryptic donor or acceptor at that offset from the annotated site, in the transcript’s direction) with its frame. The two sources are then compared (`splicevault_agreement`). A SpliceAI cryptic shift is `cryptic_confirmed` when 300K-RNA lists a cryptic site of the same kind at the same offset — the offset SpliceAI reports is genomic, so it is converted to the transcript orientation with the strand VEP supplies, falling back to magnitude when the strand is unknown — and `cryptic_unseen` otherwise, a prediction the compendium has never observed being used. A lone loss, whose outcome SpliceAI cannot state, is marked `loss_outcome_supplied`: the rank-1 SpliceVault event is the most likely consequence and supplies the frame the decomposition could not. A loss whose site type differs from the site the events are keyed to is `site_type_mismatch`, and a pure gain is `not_applicable`, because the compendium describes the use of annotated sites and is silent on sites a variant creates. The comparison is blank without SpliceVault data or without a SpliceAI event at or above the floor; blank never means disagreement. None of these columns is read by the selection gate or the tier; Step 9 appends the rank-1 event and frame to the tier reason string for the reviewer.

Wide-window rescoring (Step 5b). The precomputed SpliceAI files were generated with a 50 bp window, so a cryptic site or pseudoexon partner further than 50 bp from the variant is invisible to the screen. Every distinct variant in the calls table was therefore re-scored live with the SpliceAI model at a 4,999 bp window (the window used by the ClinGen splicing recommendations), in chunks of 1,000 variants that run as independent scheduler array tasks, with results cached per variant so a re-run of Step 5 scores nothing anew. The gene entry with the largest delta score is kept whole and decomposed exactly as above into `spliceai_wide_*` columns, together with the wide-window maximum, its difference from the precomputed maximum, and a flag marking an event more than 50 bp from the variant. These columns are evidence for review; the selection gate and the Step 9 tier continue to read the precomputed score so that results remain comparable between runs.

12. Gene-level consolidation (Step 6)

Calls were aggregated per gene symbol (falling back to the Ensembl gene identifier), counting the number of distinct probands under each model: `n_dominant` (dominant rows), `n_biallelic` (homozygous-recessive and compound-heterozygous rows), `n_xlinked` and, separately, `n_denovo`. The headline carrier count `n_carriers` is the union of the three inherited sets, and a gene is flagged recurrent at ≥ 2 distinct probands (`burden.min_carriers`). Recurrence is classified as `same_variant` when every carrier’s set of qualifying variant keys is identical (one shared allele, or one shared compound-heterozygous pair) and `distinct_variant` otherwise; same-variant recurrence is ranked below distinct-variant recurrence because a single recurrent site is as readily a founder allele or a mapping artifact as a gene-level signal.

Case-only recurrence null. For each model with at least two carriers, the probability that a random individual carries a qualifying genotype was computed under Hardy–Weinberg equilibrium from the run-oracle frequencies q_v of the gene’s observed qualifying variants, with a missing or zero frequency floored at 1×10^{-6} (`burden.absent_af_floor`): for the dominant model $p = 1 - \prod_v (1 - q_v)^2$; for the biallelic model $p = (\sum_v q_v)^2$; for the X-linked and Y-linked models $p = 1 - \prod_v (1 - q_v)$, gnomAD’s chrY frequencies being per haploid male allele. The recurrence p -value is the binomial upper tail $P(X \geq n)$ with $X \sim \text{Binomial}(N, p)$, where N is the number of resolved trios, or for the X-linked and Y-linked models the number of male probands, decided per trio by the same rule Step 5 applies (the trios file’s stated sex, else Step 0’s inference). Because every son of a carrier father carries his Y, a Y-linked recurrence across unrelated probands reflects haplogroup sharing unless the per-male frequency is very small, which is what this null measures. A carrier whose transmitting parent did not support the call (`noncarrier_parent` or `parent_gt_uninformative` in its flags) is counted like any other and also in `n_carriers_parent_unsupported`, and the audit records the genes that are recurrent only through such carriers. Every flag token carried by the emitted calls is tallied in the audit, per trio and over the cohort, so the provenance of each call class is quotable. Benjamini–Hochberg q -values are computed within each model family over the nominated genes, and an exome-wide flag is set at $p < 2.5 \times 10^{-6}$. Because the null is built only from variants observed in the cohort, it saturates on private variants and is monotone in gene size; it is reported as a rank, not as a calibrated test.

Size-normalised rank. When a per-gene mutational-target table is supplied (the gnomAD v2.1.1 constraint file, unjoined), each gene’s target is $\mu = \mu_{\text{mis}} + \mu_{\text{syn}} + \mu_{\text{lof}}$, with a missing μ_{lof} imputed as $0.0516 \times (\mu_{\text{mis}} + \mu_{\text{syn}})$ and a missing μ_{mis} or μ_{syn} yielding no target. A cohort scaling constant $C = \sum n_{\text{carriers}} / \sum \mu$ was fitted over every gene in the table, including genes with zero carriers, and each gene received `exp_carriers_mu` = $C \mu$, `carrier_excess_ratio` = $n_{\text{carriers}} / C \mu$ and `p_carrier_excess`, the Poisson upper tail $P(X \geq n_{\text{carriers}})$. With `burden.rank_by_mutational_target` (default true) recurrent genes are ordered by this size-normalised p where a target exists (`rank_basis` = `mu_normalised`), else by the smallest case-only p across models.

Constraint. A gene is labelled constrained when any of $\text{LOEUF} < 0.35$, $\text{pLI} \geq 0.90$, $\text{s_het} \geq 0.10$ or $\text{pHaplo} \geq 0.86$ holds, read from a per-gene table joined during resource preparation from gnomAD v2.1.1 (LOEUF, pLI), Zeng et al. 2024 GeneBayes s_het and Collins et al. 2022 pHaplo. Constraint is a ranking weight and never an exclusion. The column resolved for each metric is recorded in the audit.

Secondary de novo enrichment. When a Samocha-style per-gene mutation-rate table is supplied, the expected number of de novo loss-of-function plus missense variants is $2N(\mu_{\text{lof}} + \mu_{\text{mis}})$ (with μ_{lof} imputed as above when absent, and no test when μ_{mis} is absent), compared with the observed count by a Poisson upper tail; q -values are computed over the full mutation-table universe with untested genes at $p = 1$. This arm is uncalibrated against an observed synonymous rate and is reported only.

Genes are written recurrent-first, distinct-variant before same-variant, then by the rank p -value, then constrained before unconstrained, then by carrier counts and the de novo p .

13. Prioritization (Step 9)

Step 9 re-ranks the candidate calls without removing any: the number of output rows is asserted to equal the number of input rows before anything is written. Its input is Step 8's `igv/variants.tsv` when present (the only table carrying the non-human-read columns), otherwise Step 5's `candidates.calls.tsv`; a row-count mismatch between the two is fatal because Step 8 is one-to-one with Step 5.

13.1 Gene-level excess over mutational target

For each gene, n_{observed} is the number of distinct (proband, variant) observations, not rows, because a compound-heterozygous leg appears once per pair and one variant can be emitted under several modes. The expected count is $E = C \mu$, with μ as in Section 12; a gene with no gnomAD target but a CDS length receives the fallback $\log \mu = -18.4484 + 1.0570 \log(\text{CDS length})$ (an ordinary-least-squares fit over the gnomAD gene universe, roughly a $\pm 30\%$ band) and is labelled $E_{\text{source}} = \text{cds_fallback}$.

The null is a negative-binomial (NB2) model with mean E and variance $E(1 + \alpha E)$, fitted over the full gene universe (every gene in the mutational-target table with $\mu > 0$, zero-count genes included) by an iteratively trimmed procedure: $C = \sum n / \sum \mu$ over the retained genes; α by golden-section maximum likelihood on $[10^{-4}, 20]$ (or by moments, `alpha_method`); genes with $P(X \geq n) \leq 10^{-3}$ under the current fit are removed from the next fit (never from the output); the loop repeats to a fixed point (at most 20 iterations). The run halts if more than 5% of eligible genes are trimmed, which indicates a null mis-specified for the cohort rather than an exome that is 5% artifact. The upper tail $p_{\text{nb}} = P(X \geq n)$ is evaluated as the regularised incomplete beta function $I_{\{E/(E+\theta)\}}(n, \theta)$ with $\theta = 1/\alpha$, not as a complement sum, so that tails below machine epsilon remain resolvable. Benjamini–Hochberg q_{nb} is computed over the full universe, with CDS-fallback genes corrected among themselves. A Poisson tail fitted with its own trimmed scaling constant is reported alongside, and a mid- p calibration diagnostic (mean mid- p and the fraction below 0.05 and 0.001 over genes with excess ratio < 5) is written to the audit for both nulls.

13.2 Six-signal artifact panel

Each gene is scored on six binary signals, summed without weights into `corroboration_count`: cohort saturation, $n_{\text{observed}} / N \geq 0.10$; segmental duplication, the fraction of the gene covered by $\geq 98\%$ -identity duplications ≥ 0.10 (from an optional table that must be on the same coordinate build as the gnomAD v2.1.1 gene intervals, GRCh37; absent, the signal is off); membership of an artifact-prone gene family by a configurable regular-expression list (olfactory receptors, KRTAP, MUC, PRAME, NBPf, GOLGA, NPIP, HLA,

TAS2R, defensins, LCE, SPRR and histone clusters); synonymous observed/expected departure $|oe_{syn} - 1| > 0.30$; gnomAD's own constraint flag containing `mis_too_many` or `syn_outlier`; and low cumulative pLoF allele frequency, `classic_caf` at or below the 10th percentile of the candidate genes that carry a value, with a gene present in the table but lacking a value also flagged. The last is a low-information-locus signal, not a frequency signal.

13.3 Graded gene down-weight

Tiers are assigned in ascending severity and the highest matching tier wins. T1 (watch, -0.5 points): excess ratio ≥ 3 and $q_{nb} < 0.25$. T2 (-1.5): ratio ≥ 5 with at least one corroborating signal. T3 (-3.0): $q_{nb} < 0.05$ with at least one signal, or ratio ≥ 10 with at least two, or ratio ≥ 5 with saturation and at least two signals. Ratio-based rules (T2 and the two ratio limbs of T3) require $n_{observed} \geq 3$ (`excess.min_n_for_ratio_rule`); the FDR limb does not. Two ceilings are applied last: a gene in a supplied phenotype-agnostic established-gene union (for example ClinGen definitive or strong validity, dosage haploinsufficiency 3, or actionability lists) is capped at T1 and flagged `established_gene_high_excess`, which protects it from the score penalty but does not vouch for its calls; a gene whose expectation came from the CDS-length fallback is capped at T2. A ratio ≥ 10 with no corroborating signal is flagged `unexplained_excess` and carries no penalty. Two guards halt the run: an established-gene file with fewer than 1,000 genes, and T2 plus T3 covering more than 20% of distinct observations. Every tier decision is written as a human-readable `downweight_reason` that reports values and asserts no biology; a high excess ratio is a statement about calling and mapping quality, not about disease association.

13.4 Per-variant tier

Each call is assigned a molecular effect class from its consequence (canonical splice, pLoF, missense, in-frame indel, otherwise synonymous/UTR/intronic; HIGH impact of any other term is treated as pLoF) and a tier V0 to V5. V0 (molecularly benign prediction) requires impact LOW or MODIFIER with both a SpliceAI score < 0.1 and a CADD score < 15 present; a missing score can never satisfy V0. V5: a stop-gain or frameshift variant whose `nmd_status` is `triggering`, that is, one the NMD plugin assessed and did not flag as escaping nonsense-mediated decay (`nmd_escape.enabled`, default true). V4: SpliceAI ≥ 0.5 , or a HIGH-impact pLoF or canonical splice variant not promoted to V5. Canonical splice variants never reach V5, because the plugin judges the variant's own position rather than the aberrant transcript, and a pLoF whose `nmd_status` is `escaping` or `not_assessed` stays at V4: the absence of a verdict is never a promotion. pLoF confidence is UNAVAILABLE because LOFTEE is not run. V3: SpliceAI ≥ 0.2 . For missense, a single predictor speaks in a fixed order and is named in `missense_evidence_source`: REVEL ($\geq 0.773 \rightarrow V4$, $\geq 0.644 \rightarrow V3$, $\leq 0.290 \rightarrow V1$, otherwise V2); if REVEL is absent, AlphaMissense ($\geq 0.564 \rightarrow V3$, $\leq 0.34 \rightarrow V1$, otherwise V2); if both are absent, CADD $\geq 25.3 \rightarrow V3$ labelled off-label, otherwise V2. A splice rung and a missense verdict on the same variant are combined by taking the stronger tier and reporting both reasons. In-frame indels are V2; LOW/MODIFIER and synonymous/UTR/intronic variants are V1 (a discovery rank); any remaining MODERATE consequence is V2.

13.5 Additive composite and ranking

`priority_points` is an additive score in which every term is emitted as its own column (Table 2). Rarity is banded on the run oracle's value: $< 10^{-5}$ strong (+2), $< 10^{-4}$ moderate (+1.5), $< 10^{-3}$ supporting (+1), $< 10^{-2}$ permissive (+0.5), between 10^{-2} and 0.05 zero, a missing value (absent from gnomAD) +2, and ≥ 0.05 -8 . A flag marks calls whose `MAX_AF` exceeds ten times the `grpmax` proxy, indicating that a single excluded population drives the frequency. Gene constraint contributes +1 when the gene is constrained (Section 12),

multiplied by a mechanism gate on the variant tier (V0 0, V1 and V2 0.5, V3 to V5 1) and set to zero for recessive and X-linked modes, because pLoF constraint measures selection against heterozygotes. Recurrence contributes by carrier count, never by *p*-value: two distinct-variant carriers +1, three or more +2, same-variant recurrence +0.5. Quality penalties are -2 when the proband genotype fails $GQ \geq 20$, $depth \geq 10$ or the allele-balance band appropriate to its zygosity (het [0.25, 0.75]; homozygous or hemizygous ≥ 0.90 , with zygosity derived from the allele-string genotype or the mode), -3 when any screened member's non-human read fraction is flagged (Section 14.3), 0 when no member was screened, and -0.5 for every compound-heterozygous leg because the partner leg's parental QC is not carried. A biallelic call whose allele has a non-zero gnomAD homozygote count is flagged `nhomalt_recessive_conflict` and charged 0 by default. The clinical term is +4 for a ClinVar pathogenic or likely-pathogenic assertion and -4 for benign, with the positive term halved when the transferred review status is below two stars; an unavailable review status leaves full weight. Mode-of-inheritance coherence against an optional curated per-gene table contributes -1 when discordant and 0 when coherent or unknown; the penalty is suppressed, and a caveat reported instead, for a heterozygous observation in a recessive gene (the carrier-risk shape) and for a compound-heterozygous call in a gene with $CDS \geq 10,000$ bp. The gene artifact penalty is the tier penalty of Section 13.3. An optional gene-list prior (+2, scaled by an uncalibrated per-gene weight and by the same mechanism gate; off by default) enters a second total only. Two hard caps follow: a V0 variant's total is capped at 0 and a BA1 frequency caps it at -4; neither removes the row.

Two rankings are always written: `rank_agnostic` on the total without the gene-list prior and `rank_prior` with it (identical when no overlay is supplied), plus `rank_delta`. Ties are broken by molecular points, rarity points, gene artifact penalty, then genomic coordinate, so the order is reproducible. Every merged source table is also emitted verbatim under a per-source prefix (`src_mutrate_`, `src_constraint_`, `src_segdup_`, `src_moi_`, `src_burden_`, `src_prior_`) so any derived number can be re-computed from the output alone. `priority_points` is not an ACMG/AMP score and is never mapped to a pathogenicity class.

14. Review outputs (Steps 7 and 8)

14.1 Workbook

Step 7 writes an `.xlsx` workbook with an About sheet (purpose, sheet legend, run summary, the thresholds and frequency oracle read from the configuration, the VEP header line and the pipeline commit), followed by the gene consolidation table, the candidate calls, the trio resolution table, the QC report and the raw audit counts.

14.2 IGV review export

Step 8 writes a data directory for the `igv.js` trio variant-review server: `variants.tsv` with one row per candidate call (coordinates, gene symbol, consequence, impact, HGVS coding and protein descriptions, the oracle frequency as the headline frequency column, inheritance mode, origin, pair identifier, the three genotypes, the proband's GQ, depth and AB, every frequency and predictor column, ClinVar significance and stars, the non-human-read columns, and relative paths to the alignment and VCF tracks), followed by every remaining column of the calls table verbatim (the full `flags` field, the de novo tag, the Ensembl gene identifier as `gene_id`, and the `info_block`), per-trio VCF tracks, `sample_qc.tsv` from Step 0, a trio list and an empty curation state. When a sample-to-CRAM map is supplied, mini-CRAMs are sliced for each trio member around the candidate loci ($\pm 1,000$ bp, merged) with `samtools view -C --write-index`, retaining all reads by default, against the reference the CRAMs were encoded with. Slices are content-keyed on the region set and reused across runs.

14.3 Non-human read fraction (Step 8b, optional)

When a kraken2 database is supplied, the alternate-allele-supporting reads of each screened member are classified against it with `nonhuman-screen` on a temporary copy of the mini-CRAM filtered with SAM flag mask 1796 (unmapped, secondary, QC-fail and duplicate reads removed; supplementary alignments and low mapping quality retained deliberately), at kraken2 confidence 0.05 with the database memory-mapped. By default the proband is always screened and a parent only where it carries the alternate allele. Results are joined to `variants.tsv` on the zero-based variant key, yielding per-member `*_nhf` fractions with their read denominators and a derived `nhf_flag` set when any screened member has a fraction ≥ 0.5 over ≥ 5 reads. A blank denotes an unscreened member and is never read as a fraction of zero.

15. Audit and provenance

Every step appends (timestamp, step, scope, metric, value) rows to `audit/counts.tsv`, where scope is global or a trio identifier, and the orchestrator assembles `audit/summary.md`. The record covers trio resolution; the source, post-filter and union site counts; annotation coverage for each plugin and transfer; Step 3 keep and drop reasons; per-trio candidate genotype and transfer counts; Step 5's examined/skipped/called/no-row reconciliation and calls by mode; the run-level frequency oracle and per-variant basis tallies; Step 6's resolved columns, scaling constant and significance counts; and Step 9's null fit (C , α , θ , iterations, trimmed fraction, dispersion), calibration diagnostics, tier and signal tallies, and never-drop counts. Provenance columns travel with the data wherever two identical-looking values are different facts: `rarity_oracle` and `rarity_basis`; `missense_evidence_source`; `E_source` and `mu_lof_src`; `constraint_source`; `moi_coherence` and `moi_caveat`; `spliceai_status` (scored versus not covered); `nhf_status` (clean, flagged, not screened); and `clinvar_review_status` (N -star or unavailable).

16. Limitations inherent in the implementation

The screen sees single-nucleotide and small indel variation only. Loss-of-function confidence (LOFTEE) is not computed, and nonsense-mediated-decay escape is taken from the VEP NMD plugin's positional rules rather than from a transcript model; for canonical splice variants it is a position proxy and is not used for promotion. The precomputed SpliceAI set does not cover larger indels; a missing score is recorded as not covered and cannot retain a variant. SpliceVault describes how annotated sites mis-splice when disrupted and so informs loss events only; it says nothing about a site a variant creates, its events are those of a broad tissue compendium rather than the proband's tissue, and they are carried as review evidence rather than as a tier input. The case-only recurrence null is a rank, not a calibrated association test, and the de novo enrichment is uncalibrated. Under the proxy frequency arm the effective rarity stringency depends on the proband's ancestry, because the maximum is taken over ancestry groups regardless of the proband's own. The Mendelian-error and contamination statistics are computed on a capped prefix of the genome and the VCF-only contamination proxy detects only gross contamination. Genotype QC operates on posterior-derived qualities, which are not independent of the family prior the inheritance model then uses; on a male's non-pseudoautosomal chrX and on chrY, where that diploid prior is invalid, the model reads the pre-refinement likelihoods instead, which requires FORMAT/PL to be present in the callset and ignores the population prior as well. chrY is assessed as single-nucleotide and small indel variation only; AZF deletions, 45,X/46,XY mosaicism and mosaic loss of Y are copy-number events outside the screen. The variant-layer weights are reasoned rather than fitted, and the pipeline has been exercised end-to-end on generated data (Section 17) but not yet validated against truth sets.

17. Availability and testing

The code is available under the MIT licence at https://github.com/jlanej/high_priority_rare_variant; the container image is published to the GitHub Container Registry on every commit and should be pulled by digest. External resources are fetched and prepared by `scripts/prepare_resources.sh` from the pinned manifest `resources/manifest.env` (Table 1). The repository carries a pure-logic test suite (configuration validation, the annotation getters, genotype QC, the selection ladder, the whole Step 5 inheritance model against a stubbed VCF including the Y-linked model on FORMAT/PL, Step 0's chrY coverage evidence, Step 6 counting and ranking, and the Step 9 null, tiering and never-drop invariant) and an end-to-end integration test that generates a small mock genome and trios, runs the resolve step and Steps 0, 1, 3, 4, 5, 6, 8 and 9 with the real bcftools toolchain and a mocked VEP call, and asserts the resolution, funnel and calls.

Table 1. External resources and versions

Resource	Version or release	Role
Ensembl VEP and offline cache	release 115, GRCh38	consequence, IMPACT, symbol, HGVS, exon/CDS geometry, MANE, gnomAD v4.1 point AFs, ClinVar CLIN_SIG
Ensembl VEP NMD plugin	release 115 plugin set (no data file)	NMD-escape verdict behind <code>nmd_status</code> and the V5 rung
gnomAD joint sites (slim)	v4.1	<code>faf95</code> , FAF group, <code>nhomalt</code> , joint and <code>grpmax</code> AF (default frequency oracle)
CADD	v1.7, GRCh38 (whole-genome SNVs; gnomAD-genomes r4.0 indels)	functional score, third selection rung
SpliceAI precomputed raw scores	Illumina <code>genome_scores_v1.3</code> , hg38 (SNV and indel)	splice delta scores, second selection rung
REVEL	v1.3 (GRCh38-sorted)	missense tier, first predictor
AlphaMissense	hg38 release	missense tier, second predictor
SpliceVault (300K-RNA)	Ensembl FTP GRCh38 table, build 2026-02-24 (Dawes et al. 2023); optional	empirical mis-splicing events at the lost site; <code>splicevault_*</code> review columns and the agreement with the SpliceAI event
ClinVar sites VCF	2026-07-06 (pinned; monthly)	review status → gold stars
gnomAD constraint	v2.1.1 <code>lof_metrics.by_gene</code>	LOEUF, pLI; and, unjoined, $\mu_{\text{mis}}/\mu_{\text{syn}}/\mu_{\text{lof}}$, <code>oe_syn</code> , <code>classic_caf</code> , <code>constraint_flag</code> , CDS length
GeneBayes <code>s_het</code>	Zeng et al. 2024	constraint predicate
pHaplo	Collins et al. 2022	constraint predicate
kraken2 database (optional)	user-supplied, with taxonomy dumps	non-human read fraction

Resource	Version or release	Role
Reference FASTA	Ensembl release 115 GRCh38 primary assembly (or the calling reference)	normalisation, VEP, CRAM decoding

Table 2. Default thresholds and weights

Parameter	Default	Applied in
FILTER retained	PASS or .	Steps 1, 5
Excluded contigs	chrM, chrMT, M, MT	Step 1
QC sites cap	200,000	Step 0
Mendelian-error rate flag	> 0.02	Step 0
No-call rate flag	> 0.10	Step 0
chrX het fraction for male	< 0.10 over ≥ 20 sites (callset-dependent; calibrate on the parents)	Step 0
FREEMIX / VCF-proxy contamination flag	> 0.05 / > 0.02	Step 0
chrY coverage sex evidence (audit-only)	male ≥ 0.30 , female ≤ 0.10 of the father's autosome-normalised anchor depth; ≥ 50 anchors; reads at depth ≥ 3	Step 0 (Step 5 y_female_reads)
Frequency oracle	faf95 (gnomAD v4.1 joint)	Steps 3, 5, 6, 9
BA1 (drop, never rescued)	≥ 0.05	Steps 3, 9
Recessive / compound-het rarity	$< 1 \times 10^{-2}$ (high-confidence tag $< 1 \times 10^{-3}$)	Steps 3, 5
Dominant / de novo rarity	$< 1 \times 10^{-4}$	Step 5
X-linked / Y-linked (hemizygous) rarity	$< 1 \times 10^{-2}$ (high-confidence tag $< 1 \times 10^{-3}$); hemizygous de novo $< 1 \times 10^{-4}$	Step 5
Impact rung	HIGH, MODERATE	Step 3
SpliceAI rung	max $\Delta \geq 0.2$	Step 3
CADD rung	PHRED ≥ 25.3	Step 3
GQ, depth (general)	≥ 20 , ≥ 10	Steps 0, 5, 9
De novo child depth	≥ 20 (hemizygous de novo ≥ 10)	Step 5
Het allele balance	0.25–0.75	Steps 5, 9
Hom-alt allele balance	≥ 0.90	Steps 5, 9
Hom-ref allele balance; clean-parent alt reads	≤ 0.10 ; ≤ 1	Step 5
Recurrent gene	≥ 2 distinct probands	Step 6
Absent-allele floor in the recurrence null	1×10^{-6}	Step 6
Exome-wide p ; FDR	2.5×10^{-6} ; 0.05	Step 6
Constraint predicate	LOEUF < 0.35 or pLI ≥ 0.90 or s_het ≥ 0.10 or pHaplo ≥ 0.86	Steps 6, 9

Parameter	Default	Applied in
μ_{lof} imputation factor	0.0516	Steps 6, 9
Null model; trim p ; max trim fraction	NB2 (MLE α); 10^{-3} ; 0.05	Step 9
CDS fallback offset	$\log \mu = -18.4484 + 1.0570 \log L$	Step 9
Saturation; segdup; oe_syn; caf percentile	≥ 0.10 per trio; ≥ 0.10 ; $ \text{oe_syn} - 1 > 0.30$; 10th	Step 9
Tier T1 / T2 / T3	ratio ≥ 3 & $q < 0.25$ / ratio ≥ 5 & ≥ 1 signal / $q < 0.05$ & ≥ 1 , or ratio ≥ 10 & ≥ 2 , or ratio ≥ 5 & saturation & ≥ 2	Step 9
Count floor for ratio rules	$n \geq 3$	Step 9
Tier penalties T0–T3	0, -0.5 , -1.5 , -3.0	Step 9
Established-gene ceiling; CDS ceiling	T1; T2	Step 9
Min control genes; max down-weight fraction	1,000; 0.20	Step 9
SpliceAI strong / supporting / benign	≥ 0.5 / ≥ 0.2 / < 0.1	Step 9
SpliceAI event floor	0.2 (the SpliceAI rung)	Steps 5, 5b
Step 5b wide-window rescoring	enabled; 4,999 bp window; 1,000 variants per chunk	Step 5b
NMD escape grading (V5)	enabled; the VEP NMD plugin's fixed rules	Steps 2, 5, 9
CADD benign; CADD missense (off-label)	< 15 ; ≥ 25.3	Step 9
REVEL moderate / supporting / benign	≥ 0.773 / ≥ 0.644 / ≤ 0.290	Step 9
AlphaMissense supporting / benign	≥ 0.564 / ≤ 0.34	Step 9
Molecular points V0–V5	0, 0.5, 1, 2, 4, 8	Step 9
Rarity points	strong 2, moderate 1.5, supporting 1, permissive 0.5, absent 2, BA1 -8	Step 9
Constraint point; gate V0 / V1–V2 / V3–V5	$+1$; 0 / 0.5 / 1 (0 for recessive modes)	Step 9
Recurrence points	2 distinct $+1$; ≥ 3 distinct $+2$; same variant $+0.5$	Step 9
Quality points	GT fail -2 ; NHF flagged -3 ; not screened 0; comp-het partner -0.5 ; nhomalt conflict 0	Step 9
Clinical points	P/LP $+4$ ($\times 0.5$ below 2 stars); benign -4	Step 9
MOI points	discordant -1 ; unknown/coherent 0	Step 9
Gene-list prior	$+2 \times \text{weight} \times \text{gate}$; disabled by default	Step 9

Parameter	Default	Applied in
Caps	V0 total ≤ 0 ; BA1 total ≤ -4	Step 9
Long-gene CDS for comp-het caveat	$\geq 10,000$ bp	Step 9
Mini-CRAM padding	$\pm 1,000$ bp	Step 8
NHF: classify-time mask; confidence; flag fraction; min reads	1796; 0.05; ≥ 0.5 ; ≥ 5	Steps 8b, 9

Table 3. Steps, implementing code and principal outputs

All code paths are relative to the repository root: scripts under pipeline/, shared modules under src/hprv/. Outputs are relative to the run's work directory.

Step	Code (pipeline/ unless noted)	Principal outputs
resolve	resolve_trios.py;	trios.resolved.tsv,
0	src/hprv/ped.py	trio_resolution.tsv, peds/
	00_qc.py;	qc_report.tsv
	src/hprv/contamination.py,	
	genotype.py	
1	01_make_cohort_sites.sh	cohort.sites.vcf.gz
2 (2b)	02_annotate_sites.sh	cohort.sites.annotated.vcf.gz
	(02b_spliceai_backfill.sh;	
	src/hprv/spliceai_backfill.py)	
3	03_select_plausible.py;	plausible.sites.vcf.gz
	src/hprv/selection.py,	
	annotations.py	
4	04_subset_and_annotate_trios.sh	trios/<trio>.candidates.annotated.vcf.gz;
		trios.candidates.tsv
5	05_inheritance_screen.py;	candidates.calls.tsv
	src/hprv/genotype.py,	
	splice.py	
5b	05b_spliceai_rescore.sh;	candidates.calls.tsv
	src/hprv/spliceai_rescore.py,	(spliceai_wide_* appended);
	splice.py	spliceai_rescore/scores.tsv
6	06_gene_burden.py	genes.ranked.tsv
7	07_report_xlsx.py;	hprv_summary.xlsx
	src/hprv/report.py	
8 (8b)	08_igv_export.sh;	igv/variants.tsv, igv/crams/,
	src/hprv/igv.py	igv/vcfs/, igv/nhf/
9	09_prioritize.py;	variants.prioritized.tsv,
	src/hprv/prioritize.py	genes.prioritized.tsv,
		igv/variants.prioritized.tsv

Step	Code (pipeline/ unless noted)	Principal outputs
all	run_pipeline.sh, lib/common.sh; src/hprv/config.py, audit.py	audit/counts.tsv, audit/summary.md